

Arvind Murugan

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Appointments

Professor

PHYSICS + JAMES FRANCK INSTITUTE

University of Chicago

2026 -

Associate Professor

PHYSICS + JAMES FRANCK INSTITUTE

University of Chicago

2023 - 2026

Assistant Professor

PHYSICS + JAMES FRANCK INSTITUTE

University of Chicago

2015 - 2023

Postdoctoral fellow

APPLIED MATH (MICHAEL BRENNER GROUP)

Harvard University

2012 - 2015

Visiting researcher

STANISLAS LEIBLER GROUP

Rockefeller University

2009 - 2012

Member

SIMONS CENTER FOR SYSTEMS BIOLOGY

Institute for Advanced Study

2009 - 2012

Education

Princeton University

PH.D IN PHYSICS

Princeton, NJ

09/2004 - 07/2009

- Thesis: Gauge-gravity duality with renormalization group flow and reduced supersymmetry
- Advisor: Igor R. Klebanov

California Institute of Technology

B.S. IN MATHEMATICS W/ HONORS

Pasadena, CA

09/2000 - 06/2004

- Advisor: Anton Kapustin

Awards

- 2025 **Schmidt Polymath Award**, Schmidt Sciences
- 2025 **International Prize in Biophysics**, Tel Aviv
- 2023 **National Science Foundation CAREER Award**, NSF
- 2017 **Simons Investigator**, Simons Foundation
- 2010 **Addie and Harold Broitman Member in Biology**, Institute for Advanced Study
- 2004 **Princeton University Graduate Centennial Fellowship**, Princeton University

Service

2025 - 28	Advisory Board , Kavli Institute for Theoretical Physics	<i>Santa Barbara, CA</i>
2023 - 28	Deputy Director , U Chicago Center for Living Systems, NSF Physics Frontier Center	<i>Chicago, IL</i>
2026	Organizer , KITP summer workshop	<i>Santa Barbara, CA</i>
2025 - 27	Chair line , Gordon Conference Stochastic Physics in Biology	<i>Ventura, CA</i>
2024 - 25	Chair line , APS Delbruck prize committee	
2025	Planning committee , National Academies Workshop on Frontiers of Living Materials	<i>Washington D.C</i>
2025	Panelist , NSF Panel Review	<i>Alexandria, VA</i>
2025	Site Visit Panel , NSF Panel Review	<i>Alexandria, VA</i>
2024	Organizer , Aspen Institute for Physics, Winter workshop	<i>Aspen, CO</i>
2024	Panelist , NSF Panel Review	<i>Alexandria, VA</i>
2024	Selection committee , New IBD chair, UChicago	<i>Chicago, IL</i>
2022 -	Reviewing Editor , eLife	
2018 - 21	Program committee , Annual q-Bio meeting	
2017 - 23	Organizer , APS March meeting sessions	

Teaching

Ph 490	Principles of Biological Physics	<i>Fall 2015</i>
Ph 330	Mathematical Methods for Physicists	<i>Fall 2016</i>
Ph 490	Principles of Biological Physics	<i>Fall 2017</i>
Ph 330	Mathematical Methods for Physicists	<i>Fall 2017</i>
Ph 255	Biological Physics	<i>Spring 2018</i>
Ph 330	Mathematical Methods for Physicists	<i>Winter 2019</i>
Ph 121	General Physics	<i>Fall 2019</i>
Ph 330	Mathematical Methods for Physicists	<i>Winter 2020</i>
Ph 121	General Physics	<i>Fall 2020</i>
Ph 121	General Physics	<i>Fall 2021</i>
Ph 490	Biological Physics	<i>Fall 2022</i>
Ph 121	General Physics	<i>Fall 2023</i>
Ph 255	Biological Physics	<i>Winter 2023</i>
Ph 121	General Physics	<i>Fall 2024</i>
Ph 490	Biological Physics	<i>Winter 2024</i>
Ph 131	General Physics	<i>Fall 2025</i>

Support

2025 - 30	Schmidt Sciences , Schmidt Polymath Award - PI
2023 - 28	National Institute of Health , R35 award - PI
2023 - 27	National Science Foundation , Physics of Living Systems - PI
2023 - 27	National Science Foundation , CAREER award - PI
2023 - 28	National Science Foundation , Physics Frontier Center (Center for Living Systems) - Deputy Director
2023 - 26	Chan Zuckerberg Initiative , Theory Initiative - PI
2024 - 29	National Science Foundation + Simons , National Instit. for Theory and Math in Biology - Sr. Personnel
2023 - 27	Sloan Foundation , Matter-to-life - Co-PI
2022 - 27	Moore Foundation , Matter-to-life - Co-PI
2020 - 23	National Science Foundation , UChicago Materials Research Science and Eng Center - Sr. Personnel
2017 - 23	Simons Foundation , Simons Investigator, MMLS - PI

POSTDOCS

2020 -	Riccardo Ravasio	<i>Yen Fellow</i>
2024 -	Mason Rouches	<i>CLS Postdoctoral Fellow</i>
2026 -	Mats van Tongeren	
2025 -	Max Schelling	
2026 -	Cheyne Weis	
2019 - 26	Martin Falk , Currently: Faculty, UMass Amherst (2027)	<i>Schmidt Futures Fellow</i>
2023 - 26	Yoshiya Matsubara , Currently: Research A. Professor, U Tokyo (Jan 2027)	<i>Origins Initiative</i>
2023 - 24	Maryn Carlson , Currently: Faculty, U Rochester (2026)	
2018 - 23	Kabir Husain , Currently: Faculty, Physics, University College London (2024)	<i>James McDonnell Fellow</i>
2017 - 19	Zhiyue Lu , Currently: Faculty, Chemistry, UNC Chapel Hill (2024)	

GRADUATE STUDENTS

2025 -	Serena Debesai	<i>NSF GRFP</i>
2025 -	Ruby Wen	
2024 -	Rio Ondo	<i>PME</i>
2024 -	Kristina Trifonova	<i>NSF GRFP</i>
2023 -	Darren Liu	
2026 -	Kristen Vosberg	
2026 -	Vamsi Varanasi	
2022 -	LaNijah Flagg	
2020 -	Christopher Russo	
2022 - 26	Milena Chakraverti-Wuerthwein	<i>Hertz Fellow</i>
2018 - 25	Eric Rouviere , Currently: Postdoc, JHU (2025)	
2020 - 24	Rudy Mendez Reina	
2017 - 22	Vedant Sachdeva , Currently: Evozyne (biotech startup) (2023)	<i>Harper Dissertation Fellow</i>
2016 - 21	Jackson O'Brien , Currently: Lincoln Park Capital (biotech VC) (2023)	
2016 - 21	Weerapat Pittayakanchit , Currently: Krungthai Bank, Thailand (2023)	
2018 - 20	Chukowunso Arinze , Currently: Kaoshi (founder) (2022)	
2016 - 19	Menachem Stern , Currently: Faculty, AMOLF (2024)	<i>APS GSNP award, Bloomenthal prize</i>

UNDERGRADUATES

HuiYi Mei	<i>2025 - 2026</i>
Leon Zhou	<i>2022 - 2026</i>
Aidan Etterer	<i>2024 - 2025</i>
Michele Peruzzo	<i>2022 - 2025</i>
Sumana Turimella	<i>2022 - 2024</i>
Sedona Kessler , Now: Undergrad, Harvard	<i>2023 - 2024</i>
Shiv Seshan , Now: Grad Student, Caltech (2025)	<i>2023 - 2024</i>
Finn Roach	<i>2021 - 2022</i>
Sophia Smith , Now: Grad Student, UT Austin Applied Math (2022)	<i>2020 - 2021</i>
Sophia Chiang	<i>2020 - 2021</i>
Viraaj Jayaram , Now: Grad Student, Yale Physics (2022)	<i>2017 - 2019</i>
Leron Perez , Now: Grad Student, Stanford (2022)	<i>2018 - 2019</i>
Jon Kutasov , Now: Optiver (fintech) (2021)	<i>2019 - 2019</i>
Amber Bourdier , Now: ESILV, Paris (2019)	<i>2017 - 2018</i>
Ryan Thiermann , Now: Grad Student, UC San Diego Physics (2022)	<i>2015 - 2017</i>
Weishun Zhong , Now: Grad Student, MIT Physics (2022)	<i>2016 - 2017</i>
Alexandra Carruthers Ferrero , Now: Grad Student, Univ Puerto Rico (2018)	<i>2016 - 2016</i>

Articles

1. Alston, H., Rouches, M., Murugan, A., Walczak, A. M. & Mora, T. Theoretical limits for sensing through phase separation. *Proceedings of the National Academy of Sciences* **123**, e2520040123 (2026).
2. Chakraverti-Wuerthwein, M. S., Matsubara, Y. J., Roach, F. D., Narla, A. V., Hwa, T. & Murugan, A. Minimizing co-growth as a broad predictor of community robustness. *bioRxiv* (2026).
3. Ravasio, R., Husain, K., Evans, C. G., Phillips, R., Ribezzi-Crivellari, M., Szostak, J. W. & Murugan, A. Evolution of error correction through a need for speed. *Science* **391**, 818–824 (2026).
4. Crocker, K., Skwara, A., Kannan, R., Murugan, A. & Kuehn, S. Timescale of environmental change modulates metabolic guild cohesion in microbial communities. *The ISME Journal* **19**, wrf186 (2025).
5. Dhanuka, A., Flamholz, A. I., Murugan, A. & Goyal, A. Ecosystems as adaptive living circuits. *bioRxiv*, 2025.06. 27.661910 (2025). Under revisions at Nature Communications.
6. Falk, M. J., Strupp, A. T., Scellier, B. & Murugan, A. Temporal Contrastive Learning through implicit non-equilibrium memory. *Nature Communications* **16**, 2163 (2025).
7. Ferretti, F., Kardar, M. & Murugan, A. Learning to generalize in evolution through annealed population heterogeneity. *arXiv preprint arXiv:2510.24602* (2025). Under revisions at PNAS.
8. Floyd, C., Dinner, A. R., Murugan, A. & Vaikuntanathan, S. Limits on the computational expressivity of non-equilibrium biophysical processes. *Nature Communications* **16**, 7184 (2025).
9. Murugan, A., Zwicker, D., Lorenz, C. & Dufresne, E. R. Could Living Cells Use Phase Transitions to Process Information? *arXiv preprint arXiv:2507.23384* (2025). Accepted at Physical Review X.
10. Narla, A. V., Hwa, T. & Murugan, A. Dynamic coexistence driven by physiological transitions in microbial communities. *Proceedings of the National Academy of Sciences* **122**, e2405527122 (2025).
11. Russo, C. J., Husain, K., Ranganathan, R., Pincus, D. & Murugan, A. Simple biological controllers drive the evolution of soft modes. *arXiv preprint arXiv:2507.11973* (2025). Accepted at PNAS.
12. Trifonova, K., Falk, M. J., Rouches, M., Vaikuntanathan, S., Elowitz, M. B. & Murugan, A. Trainable computation in molecular networks. *bioRxiv*, 2025.12. 28.696421 (2025).
13. Zentner, A., Halingstad, E. V., Chalk, C., Brenner, M. P., Murugan, A., Winfree, E. & Shrinivas, K. Information processing driven by multicomponent surface condensates. *arXiv preprint arXiv:2509.08100* (2025). Accepted at PNAS.
14. Chalk, C., Buse, S., Shrinivas, K., Murugan, A. & Winfree, E. *Learning and Inference in a Lattice Model of Multicomponent Condensates* 5: 1–5: 24 DNA 30 (Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2024).
15. Evans, C. G., O'Brien, J., Winfree, E. & Murugan, A. Pattern recognition in the nucleation kinetics of non-equilibrium self-assembly. *Nature* **625**, 500–507 (2024).
16. Falk, M. J., Roach, F. D., Gilpin, W. & Murugan, A. Curiosity-driven search for novel nonequilibrium behaviors. *Physical Review Research* **6**, 033052 (2024).
17. Husain, K., Sachdeva, V., Ravasio, R., Peruzzo, M., Liu, W., Good, B. H. & Murugan, A. Direct and indirect selection in a proofreading polymerase. *bioRxiv* (2024). Under revisions at Nature Communications.
18. Jaeger, H. M., Murugan, A. & Nagel, S. R. *Training physical matter to matter* **34**, 6695–6701 (Royal Society of Chemistry, 2024).
19. Mandal, R., Huang, R., Fruchart, M., Moerman, P. G., Vaikuntanathan, S., Murugan, A. & Vitelli, V. Learning dynamical behaviors in physical systems. *arXiv preprint arXiv:2406.07856* (2024).
20. Ravasio, R., Husain, K., Evans, C. G., Phillips, R., Ribezzi, M., Szostak, J. W. & Murugan, A. A minimal scenario for the origin of non-equilibrium order. *arXiv preprint arXiv:2405.10911* (2024).
21. Russo, C. J., Husain, K. & Murugan, A. *Soft Modes as a Predictive Framework for Low-Dimensional Biological Systems Across Scales* (Annual Reviews, 2024).
22. Arinze, C., Stern, M., Nagel, S. R. & Murugan, A. Learning to self-fold at a bifurcation. *Physical Review E* **107**, 025001 (2023).
23. Devany, J., Falk, M. J., Holt, L. J., Murugan, A. & Gardel, M. L. Epithelial tissue confinement inhibits cell growth and leads to volume-reducing divisions. *Developmental cell* **58**, 1462–1476. e8 (2023).
24. Falk, M., Strupp, A., Scellier, B. & Murugan, A. Contrastive learning through non-equilibrium memory. *arXiv preprint arXiv:2312.17723* (2023).
25. Falk, M. J., Wu, J., Matthews, A., Sachdeva, V., Pashine, N., Gardel, M. L., Nagel, S. R. & Murugan, A. Learning to learn by using nonequilibrium training protocols for adaptable materials. *Proceedings of the National Academy of Sciences* **120**, e2219558120 (2023).
26. Goyal, A., Flamholz, A. I., Petroff, A. P. & Murugan, A. Closed ecosystems extract energy through self-organized nutrient cycles. *arXiv preprint arXiv:2305.19102* (2023).
27. Stern, M. & Murugan, A. Learning without neurons in physical systems. *Annual Review of Condensed Matter Physics* **14**, 417–441 (2023).
28. Apte, A., Marwaha, K. & Murugan, A. Non-Convex Optimization by Hamiltonian Alternation. *arXiv preprint arXiv:2206.14072* (2022).
29. Falk, M. J., Roach, F., Gilpin, W. & Murugan, A. Curiosity search for non-equilibrium behaviors in a dynamically learned order parameter space. *arXiv preprint arXiv:2211.02589* (2022).
30. Schaffter, S. W., Chen, K.-L., O'Brien, J., Noble, M., Murugan, A. & Schulman, R. Standardized excitable elements for scalable engineering of far-from-equilibrium chemical networks. *Nature Chemistry* **14**, 1224–1232 (2022).
31. Su, C. J., Murugan, A., Linton, J. M., Yeluri, A., Bois, J., Klumpe, H., Langley, M. A., Antebi, Y. E. & Elowitz, M. B. Ligand-receptor promiscuity enables cellular addressing. *Cell systems* **13**, 408–425 (2022).
32. Falk, M. J., Alizadehyazdi, V., Jaeger, H. & Murugan, A. Learning to control active matter. *Physical Review Research* **3**, 033291 (2021).

33. Murugan, A., Husain, K., Rust, M. J., Hepler, C., Bass, J., Pietsch, J. M., Swain, P. S., Jena, S. G., Toettcher, J. E., Chakraborty, A. K., *et al.* Roadmap on biology in time varying environments. *Physical biology* **18**, 041502 (2021).
34. Son, M., Wang, A. G., Tu, H.-L., Metzger, M. O., Patel, P., Husain, K., Lin, J., Murugan, A., Hoffmann, A. & Tay, S. NF- κ B responds to absolute differences in cytokine concentrations. *Science Signaling* **14**, eaaz4382 (2021).
35. Galst'yan, V., Husain, K., Xiao, F., Murugan, A. & Phillips, R. Proofreading through spatial gradients. *Elife* **9**, e60415 (2020).
36. Husain, K. & Murugan, A. Physical constraints on epistasis. *Molecular Biology and Evolution* **37**, 2865–2874 (2020).
37. Sachdeva, V., Husain, K., Sheng, J., Wang, S. & Murugan, A. Tuning environmental timescales to evolve and maintain generalists. *Proceedings of the National Academy of Sciences* **117**, 12693–12699 (2020).
38. Stern, M., Arinze, C., Perez, L., Palmer, S. E. & Murugan, A. Supervised learning through physical changes in a mechanical system. *Proceedings of the National Academy of Sciences* **117**, 14843–14850 (2020).
39. Stern, M., Pinson, M. B. & Murugan, A. Continual learning of multiple memories in mechanical networks. *Physical Review X* **10**, 031044 (2020).
40. Zhong, W., Lu, Z., Schwab, D. J. & Murugan, A. Nonequilibrium statistical mechanics of continuous attractors. *Neural Computation* **32**, 1033–1068 (2020).
41. Husain, K., Pittayakanchit, W., Pattanayak, G., Rust, M. J. & Murugan, A. Kalman-like Self-Tuned sensitivity in biophysical sensing. *Cell systems* **9**, 459–465 (2019).
42. Murugan, A. & Jaeger, H. M. Bioinspired nonequilibrium search for novel materials. *MRS Bulletin* **44**, 96–105 (2019).
43. O'Brien, J. & Murugan, A. Temporal pattern recognition through analog molecular computation. *ACS Synthetic Biology* **8**, 826–832 (2019).
44. Chew, J., Lypunskiy, E., Lin, J., Murugan, A. & Rust, M. J. High protein copy number is required to suppress stochasticity in the cyanobacterial circadian clock. *Nature communications* **9**, 3004 (2018).
45. Pittayakanchit, W., Lu, Z., Chew, J., Rust, M. J. & Murugan, A. Biophysical clocks face a trade-off between internal and external noise resistance. *Elife* **7**, e37624 (2018).
46. Stern, M., Jayaram, V. & Murugan, A. Shaping the topology of folding pathways in mechanical systems. *Nature communications* **9**, 4303 (2018).
47. Thiermann, R., Sweeney, A. & Murugan, A. Information content of downwelling skylight for non-imaging visual systems. *bioRxiv*, 408989 (2018).
48. Murugan, A. & Vaikuntanathan, S. Topologically protected modes in non-equilibrium stochastic systems. *Nature communications* **8**, 13881 (2017).
49. Pinson, M. B., Stern, M., Carruthers Ferrero, A., Witten, T. A., Chen, E. & Murugan, A. Self-folding origami at any energy scale. *Nature communications* **8**, 15477 (2017).
50. Stern, M., Pinson, M. B. & Murugan, A. The complexity of folding self-folding origami. *Physical Review X* **7**, 041070 (2017).
51. Zhong, W., Schwab, D. J. & Murugan, A. Associative pattern recognition through macro-molecular self-assembly. *Journal of Statistical Physics* **167**, 806–826 (2017).
52. Huntley, M. H., Murugan, A. & Brenner, M. P. Information capacity of specific interactions. *Proceedings of the National Academy of Sciences* **113**, 5841–5846 (2016).
53. Murugan, A. Renormalization group flows in gauge-gravity duality. *arXiv preprint arXiv:1610.03166* (2016).
54. Murugan, A. & Vaikuntanathan, S. Biological implications of dynamical phases in non-equilibrium networks. *Journal of Statistical Physics* **162**, 1183–1202 (2016).
55. Zwicker, D., Murugan, A. & Brenner, M. P. Receptor arrays optimized for natural odor statistics. *Proceedings of the National Academy of Sciences* **113**, 5570–5575 (2016).
56. Murugan, A., Zeravcic, Z., Brenner, M. P. & Leibler, S. Multifarious assembly mixtures: Systems allowing retrieval of diverse stored structures. *Proceedings of the National Academy of Sciences* **112**, 54–59 (2015).
57. Murugan, A., Zou, J. & Brenner, M. P. Undesired usage and the robust self-assembly of heterogeneous structures. *Nature communications* **6**, 6203 (2015).
58. Murugan, A., Huse, D. A. & Leibler, S. Discriminatory proofreading regimes in nonequilibrium systems. *Physical Review X* **4**, 021016 (2014).
59. Murugan, A., Huse, D. A. & Leibler, S. Speed, dissipation, and error in kinetic proofreading. *Proceedings of the National Academy of Sciences* **109**, 12034–12039 (2012).
60. Klebanov, I. R., Klose, T. & Murugan, A. AdS₄/CFT₃ squashed, stretched and warped. *Journal of High Energy Physics* **2009**, 140 (2009).
61. Klebanov, I. R., Kutasov, D. & Murugan, A. Entanglement as a probe of confinement. *Nuclear physics B* **796**, 274–293 (2008).
62. Klebanov, I., Murugan, A., Rodriguez-Gomez, D. & Ward, J. Goldstone bosons and global strings in a warped resolved conifold. *Journal of High Energy Physics* **2008**, 090 (2008).
63. Klebanov, I. R. & Murugan, A. Gauge/gravity duality and warped resolved conifold. *Journal of High Energy Physics* **2007**, 042 (2007).
64. Baumann, D., Dymarsky, A., Klebanov, I. R., Maldacena, J., McAllister, L. & Murugan, A. On D3-brane potentials in compactifications with fluxes and wrapped D-branes. *Journal of High Energy Physics* **2006**, 031 (2006).
65. Kapustin, A. & Murugan, A. Fatgraph expansion for noncritical superstrings. *arXiv preprint hep-th/0404238* (2004).

Invited talks

Sep 26 **Cornell**, LASSP/AEP Seminar
 Sep 26 **Annual Origins Federation meeting**, Life in the Universe
 Jul 26 **KITP**, Learning with a brain

Ithaca, NY
 Zurich, Switzerland
 Santa Barbara, CA

Jun 26	Gulbenkian , Molecular and Biophysical Principles in Cellular Evolution	Lisbon, Portugal
Apr 26	Brown , Physics Colloquium	
Mar 26	APS March meeting , AI + Soft Matter Invited Session	Denver, CO
Mar 26	MIT , Biophysics seminar	Cambridge, MA
Feb 26	Georgetown	Washington, DC
Feb 26	Yale , Physics Colloquium	New Haven, CT
Jan 26	Les Houches , Computing with matter	Les Houches, France
Jan 26	Weizmann Institute	Jerusalem, Israel
Jan 26	Tel Aviv University , International Biophysics Prize seminar	Tel Aviv, Israel
Jan 26	Hebrew University	Jerusalem, Israel
Dec 25	American Society for Cell Biology , Cell Behavior and Cognition	Philadelphia, PA
Dec 25	Laufer Center for Physical and Quantitative Biology	Stony Brook, NY
Sep 25	DARPA , Complexity Roundtable	Washington, DC
Jul 25	Molecular Origins of Life 2025	Online
Jul 25	Nordita , Eco-evolutionary Processes in Context	Stockholm, Sweden
Jun 25	ESPCI , Chaire Joliot seminar	Paris, France
Jun 25	Beg Rohu Summer school , Learning with Machines, Physics and Minds	Beg Rohu, France
May 25	Aspen Colloquium , Beyond von Neumann computing	Aspen, CO
May 25	NITMB , Origin of Life and Self-Organized Complexity	Chicago, IL
May 25	EMBL , Theory Transversal Theme retreat	Heidelberg, Germany
May 25	EMBL Symposium , Theory and concepts in biology	Heidelberg, Germany
Apr 25	Johns Hopkins , Materials Science Seminar	Baltimore, MD
Jan 25	UC Berkeley , Annual Berkeley Stat Mech Meeting	Berkeley, CA
Jan 25	National Institute for Theory and Math in Biology , Biological systems that learn	Chicago, IL
Nov 24	Cornell , Soft Matter Hub	Online
Aug 24	Aspen Summer Workshop Data in Biology , Biological Data Analysis	Aspen, CO
Jul 24	MOLINFO 2024 , Molecular Information	Munich, Germany
Jul 24	University of Tokyo , Workshop on Frontiers in Biophysics	Tokyo, Japan
Jun 24	IUPAB congress 2024 , Theory in Biology symposium	Kyoto, Japan
Jun 24	Gordon Research Conference , Systems Chemistry	Portland, ME
Jun 24	University of Pennsylvania , Center for Soft and Living Matter kickoff	Philadelphia, PA
May 24	MIT , Applied Math Seminar	Cambridge, MA
Apr 24	1st Solvay conference in Biology , The organisation and dynamics of biological computation	Brussels, Belgium
Apr 24	Simons Foundation , NITMB Annual Meeting	New York, NY
Mar 24	APS March meeting , Invited talk	Minneapolis, MN
Jan 24	Aspen Institute for Physics , Computation in Physical systems	Aspen, CO
Oct 23	29th Solvay conference on Physics , Structure and Dynamics of Disordered Systems	Brussels, Belgium
Sep 23	American Academy of Arts and Sciences , Life in the Universe II	Boston, MA
Sep 23	University of California, Merced (Remote) , Physics colloquium	Merced, CA
Jul 23	University of Michigan , ICAM Complex Mechanical Metamaterials workshop	Ann Arbor, MI
Jun 23	Gordon Conference on Molecular Mechanisms of Evolution	Stonehill College, MA
Jun 23	Northwestern University , qBio Symposium	Chicago, IL
Mar 23	APS March meeting , Statistical physics Tutorial	Las Vegas, NV
Mar 23	NYU , Physics colloquium	NYC, NY
Feb 23	Harvard Medical School , Theory lunch	Boston, MA
Jan 23	Gordon Research Conference , Stochastic Processes in Biology	Ventura, CA
Nov 22	AMOLF , Physics Colloquium	Amsterdam, NL
Nov 22	AMOLF , Workshop on Information Processing	Amsterdam, NL
Nov 22	TU Eindhoven , Physics/Chemistry Colloquium	Eindhoven, NL
Oct 22	Stanford , Coherent Network Computing	Palo Alto, CA
Sep 22	Cargese Summer School , Summer school	Corsica, France
May 22	Simons Center for Geometry & Physics , Geometry, Topology, Symmetry in Soft & Living Matter	Stony Brook, NY
May 22	Johns Hopkins , Biophysics seminar	Baltimore, MD
Apr 22	Harvard University , Widely Applied Mathematics	Cambridge, MA
Apr 22	Illinois Institute of Technology	Chicago, IL
Mar 22	APS March Meeting , Invited talk	Chicago, IL
Sep 21	University of Michigan , Physics seminar	Ann Arbor, MI
May 21	Cold Spring Harbor (remote)	Cold Spring Harbor, NY
May 21	University of Amsterdam (remote)	Amsterdam, NL

Apr 21	Biological Physics/Physical Biology series (Virtual)	Zoom
Jan 21	qEvo 2021	Zoom
Dec 20	Max Planck Institute (remote)	Gottingen, Germany
Nov 20	Imperial College , Physics of life series	London, UK
Jun 20	Tel Aviv University , Workshop on Pathways, Sequence and Memory	Tel Aviv, Israel
Feb 20	Banff International Research Station , Workshop on gene regulation	Banff, Canada
Jan 20	Massachusetts Institute of Technology , Physics seminar	Cambridge, MA
Jan 20	EPFL , CECAM workshop	Lausanne, Switzerland
Oct 19	University of California, Los Angeles , Physics seminar	Los Angeles, CA
Sep 19	National Science Foundation , MRSEC Directors meeting	Alexandria, VA
Sep 19	University of Pennsylvania , Physics seminar	Philadelphia, PA
Jul 19	Aspen Center for Physics , Information processing in cells	Aspen, CO
Jun 19	Frontiers of Biophysics	Paros, Greece
Mar 19	Chan-Zuckerberg Biohub , Theory in Biology day	San Francisco, CA
Jan 19	Gordon Research Conference , Stochastic Physics and Biology	Ventura, CA
Nov 18	University of Illinois , Physics seminar	Urbana-Champaign, IL
Oct 18	Peking University , Center for Quantitative Biology	Beijing, China
Oct 18	DNA 24 , Plenary Talk	Dajin, China
Oct 18	Penn State , Physics seminar	College Station, PA
Apr 18	Emory University , Physics Colloquium	Atlanta, GA
Apr 18	Simons Foundation , Theory in Biology conference	New York, NY
Apr 18	Rice University , Physics seminar	Houston, TX
Mar 18	APS March Meeting , Invited talk	Los Angeles, CA
Feb 18	KITP , Memories workshop	Santa Barbara, CA
Nov 17	Princeton Center for Theoretical Science , Workshop on frustration	Princeton, NJ
Oct 17	MIT , Physics seminar	Cambridge, MA
Oct 17	University of Texas Southwestern , Systems Biology Seminar	Dallas, TX
Sep 17	Washington University in St. Louis , Physics seminar	St Louis, MO
Jul 17	QBio workshop	New Brunswick, NJ
May 17	Los Alamos National Lab , CNLS Annual workshop	Santa Fe, NM
Mar 17	Harvard University , Widely Applied Math seminar	Cambridge, MA
Jan 17	Northwestern University , Mechanical Engineering,	Chicago, IL
May 16	Harvard University , Workshop on Aging and Failure in biology	Cambridge, MA
Apr 16	Foundations of Nanoscience (FNANO) , Invited speaker	Snowbird, UT
Mar 16	University of Illinois Urbana-Champaign , Urbana cond-mat Symposium	Urbana, IL
Sep 15	Santa Fe Institute , Kinetic Networks workshop	Santa Fe, NM
Apr 15	Rockefeller University , Center for Studies in Physics and Biology	New York, NY
Mar 15	University of Colorado , Physics Colloquium	Boulder, CO
Mar 15	University of Colorado , Physics seminar	Boulder, CO
Mar 15	APS March Meeting , Invited Talk	San Antonio, TX
Mar 15	University of Massachusetts , Physics seminar	Amherst, MA
Feb 15	University of San Diego , Physics seminar	San Diego, CA
Feb 15	MIT , Physics seminar	Cambridge, MA
Feb 15	Aspen Institute for , Unified Concepts in Glass Physics VI,	Aspen, CO
Jan 15	Cornell , LASSP Applied Physics seminar	Ithaca, NY
Jan 15	University of Chicago , James Franck Institute,	Chicago, IL
Dec 14	Princeton University , Department of Chemistry,	Princeton, NJ
Jul 14	Universite Joseph Fourier	Grenoble, France
Jun 14	University of Nice , Laboratory of Condensed Matter Physics seminar	Nice, France
Jun 14	ESPCI	Paris, France
Jun 14	Brandeis , Physics Department	Waltham, MA
May 14	University of Chicago , Computations in Science Seminar	Chicago, IL
Apr 14	University of Maryland , Institute for Physical Science and Technology,	College Park, MD
Jan 14	Northwestern , Applied Math Colloquium	Evanston, IL
Dec 13	Boston Physical Biology Hangout	Cambridge, MA
Nov 13	Harvard , Bauer Forum	Cambridge, MA
Dec 08	Perimeter Institute for Theoretical Physics , Young Researchers Conference	Waterloo, ON
May 08	Institute for Nuclear Theory , String Theory Methods in the Real World,	Seattle, WA
Apr 08	Princeton University , High-energy theory group seminar	Princeton, NJ

Apr 08 **Cornell University**, High-energy theory group seminar
Nov 07 **University of Chicago**, High-energy theory group seminar

Ithaca, NY
Chicago, IL